

PLO Camp

Faraz

1 Problems

This is a decently large collection of problems that cover most topics that would come up in LOs. Problems are selected based on their suitability. We will discuss a subset of these problems at Camp. Feel free to solve things beforehand because you probably won't have enough time at Camp. Huge huge thanks to Deeraj Pothapragada, Matei Chirila and Panawat Tiacharoen without whom this document wouldn't exist. Shoot us up for questions if you have any. Happy problem solving!

Problem 1. Fijian (2015, ROL)

Here are some words in Fijian, with the primary and secondary stresses marked: láko, paràimarí:, tálo, βináka, kilá:, nrè:nré:, atómi, perèsiténdi, mìnisi:terí:, mbàsikètepólo, mbè:léti, seŋái, taràusése, parò:karámu, mì:sìningáni, ndàirèkitá:

- (a) Mark the primary and secondary stresses in the following words:
mbelembo:tomu, mbasa:, ndikonesi, ndoketa:, palasita:, terenisisita:

△ : marks that the preceding vowels long, ' marks primary stress and, ` marks secondary stress.

Problem 2. Tabla (2021, KrLO)

Below are some words in the Tabla language, broken into syllables with the stressed syllable capitalized. ə is the 'a' sound in "about"

Tabla	Translation	Tabla	Translation
A-ka	every	pan-KO-ru	floor
E-nin	afraid	an-SI-ran	bow
DAI-tə	moment	na-RI-yan	rainbow
A-bə-re	foreign	ki-NAU-wai	middle-aged
Bə-rən-ka	head	ba-rə-MA-rə	I will throw
SUN-kə-nun	mosquito	ti-de-rə-PO-də	suddenly
Mə-rə-wai	tart	nə-sə-REI-bei	I give it
ti-NI-mi	near	i-Tə-wə-rə	eaten
eu-TO-na	lizard	a-NA-də-na	day

- (a) Find the stress for the following words:
a-rən-ka parrot, ba-ro seagull, də-rei-mo-te ear

Problem 3. Ligurian (2020, UKLO)

Below are some words written in Ligurian along with their English translations. The stress is indicated by the mark ˘ above the vowel. Two of the words have their stress marked on the wrong syllable.

ɔ:ʒél:i	<i>birds</i>
sité:	<i>city</i>
pásta	<i>pasta</i>
skwád:ra	<i>team</i>
vjú:vet:a	<i>purple</i>
damáskina	<i>pear</i>
nóstru	<i>our</i>
venín	<i>poison</i>
du:seménte	<i>sweetly</i>
kutél:u	<i>knife</i>
kúm:e	<i>how</i>
mejʒín:a	<i>medicine</i>
dát:ɔ	<i>date (fruit)</i>
pe:tená:	<i>to comb</i>
ba:ʒú	<i>kiss</i>
májstra	<i>teacher</i>
pú:vje	<i>dust</i>
rám:u	<i>copper</i>
taramót:u	<i>earthquake</i>
agýs:u	<i>sharp</i>
agysá:	<i>to sharpen</i>
béstja	<i>beast</i>

(a) Write the correct stresses for the two words.

(b) Add accents to **bolac:ɔ** *beast*, **se:i** *six*, **furmine** *lightning*, **abitud:ine** *habit*, **acordjo** *agreement*, **ε:gwa** *water*

⚠ In Ligurian, both consonants and vowels can be long (these are marked with the sign : placed after the sound); ʒ, j, ŋ and w are consonants; ɔ, y and ε are vowels.

Problem 4. Persian (2022, CLO)

Persian is an Iranian language from the Indo-European language family. It is an official language in Iran, Afghanistan and Tajikistan. Poetry forms a significant part of the rich Persian literature. Your task will be to understand the meters used in

their works. Below you will find two verses written in motaqāreb. The first is from Firdausí's Shahnameh, the second from Sa'dí's Gulistan.

be nāmē Xodāvande jān ō xerad
našāyad ke nāmat nahand ādamī

If we were to divide these verses into syllables, it would look like this:

be/nā/mē/Xo/dā/van/de/jā/nō/xe/rad
na/šā/yad/ke/nā/mat/na/han/dā/da/mī

- (a) The following six verses were written by Sa'di. The first four come from Gulistan, the other two from Bustan. All verses are written in motaqāreb. Complete the verses with the words in brackets. For five verses there is unambiguous "rhythmically correct" addition. The remaining verse can be completed in two ways, both additions will be considered correct.

__ (1) __ peykarand (a'zāye, yek, banī'Ādam)
ke __ (2) __ gowharand (dar, yek, ze, 'āfarīneš)
čo __ (3) __ rūzgār (āvarad, be, dard, 'ozvī)
degar __ (4) __ qarār (namānad, 'ozvhārā)
be __ (5) __ āfarīd (jān, ke, nāmē, Xodāyī)
soxan __ (6) __ āfarīd (andar, goftan, zabān)

△ marks that a vowel is long. ' is a consonant

Problem 5. Gusii (2023, KrLO)

Below are some verb forms in the Gusii language

Gusii	Translation	Gusii	Translation
címbú	Leave!	ò kındílá	He pressed
í címbú	I left	ò kındílà lè	He should'nt press
íí címbú	I used to leave	láá kòl	You _{pl} used to drink
láá címbù	You _{pl} used to leave	à hòlíhólí	You _{sg} put it on
là tómbòlà lè	You _{pl} shouldn't say	í hòlíhólí	I put it on
í pèndú	I bought it	dél	Fall down!
à pèndú	You _{sg} bought it	ò dël	He fell down
í pèndù lè	I shouldn't buy it	íí dël	I used to fall down
óó pèndù	He used to buy	ò sùkásúká	He rinsed his mouth

- (a) Write in English:

1. kındílá
2. ò kòl lè

3. à tòmbóla

(b) Write in Gusii:

4. *he drank*
5. *you_{pl} shouldn't put it on*
6. *I fell down*
7. *I shouldn't fall down*
8. *Rinse your mouth!*
9. *you_{sg} used to rinse your mouth*
10. *I used to say*

△, ^, ˇ, and ^ marked above vowels denote high, low, rising, and falling tones respectively.

Problem 6. Dutch (2012, UKLO, 1 pg)

Here are some verbs in Dutch and their participle forms:

Verb	Participle
slibben	geslibd
klagen	geklaagd
branden	gebrand
weren	geweerd
tochten	getocht
tellen	geteld
raken	geraakt
lijmen	gelijmd
kunnen	gekund
vertellen	verteld
telen	geteeld
verhoren	verhoord
trouwen	getrouwd
schaven	geschaafd
razen	geraasd
prijzen	geprijsd
lappen	gelapt
smakken	gesmakt
praten	gepraat
fietsten	gefietst
boffen	geboft

- (a) What are the past participles of the following verbs: **delen**, **horen**, **tappen**, **verhuizen**, **landen**, **kloppen**, **mokken**, **roken**, **rotten**, **tobben**.

Problem 7. Haitian Creole (2012, NACLO, 1 pg)

Haitian Creole is a language spoken by 10 million people in Haiti. It is based on French and some African languages. The combinations **an**, **en**, and **on** denote nasal vowels (partially produced through the nose). The table below gives you nouns in Haitian Creole with definite articles and the corresponding English translations:

Noun + Article	Translation
kòl la	<i>the necktie</i>
jaden an	<i>the garden</i>
lanp lan	<i>the lamp</i>
kouto yo	<i>the knives</i>
mayi a	<i>the maize</i>
lanmou an	<i>the love</i>
ri a	<i>the street</i>
pant lan	<i>the slope</i>
manman an	<i>the mother</i>
kabann nan	<i>the bed</i>
sant yo	<i>the odors</i>
lèt la	<i>the letters</i>
sans lan	<i>the sense</i>
telefòn nan	<i>the telephone</i>
kò a	<i>the body</i>
mi an	<i>the wall</i>
fanm nan	<i>the woman</i>
joumou an	<i>the pumpkin</i>
ban an	<i>the beach / the ban</i>
kay la	<i>the house</i>
fanmi an	<i>the family</i>
pon an	<i>the bridge</i>
liv yo	<i>the books</i>
dam nan	<i>the lady</i>

- (a) Provide the articles for the following nouns: **lalin** moon, **pitimi** millet, **muzik** music, **maldamou** love-sickness, **kap** kite, **radio** radio, **vant** belly, **tan** time
- (b) Translate the following into Haitian Creole: *the knife, the bodies, the odor, the book*
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Problem 8. Turkmen (2021, Polish LO, 1 pg)

Here are some Turkmen verbs written in the Turkmen alphabet in two forms and the meaning in English, with some forms missing.

Form A	Form B	Meaning	Form A	Form B	Meaning
diýil	diýen	to speak	geplen	geplän	to converse
işlen	işlän	to work	ýörel	ýörän	to go
görüň	gören	to look	guradyl	guradan	to dry
beril	beren	to give	sanal	sanan	to count
okal	okan	to read/learn	__(1)__	bezän	to decorate
hapalan	hapalan	to make dirty	__(2)__	gelen	to arrive
ýuwul	ýuwan	to wash	__(3)__	söýen	to love
ýazyl	ýazan	to write	__(4)__	iberen	to send
ylgal	ylgan	to run	__(5)__	çüýlän	to hammer in
gurul	guran	to build	bölün	__(6)__	to split
alyn	alan	to take	goral	__(7)__	to protect
sözlen	sözlän	to say	salyn	__(8)__	to place
söweşil	söweşen	to fight	çeynel	__(9)__	to chew
geplen	geplän	to converse	süzül	__(10)__	to filter

(a) Fill in blanks 1-10.

Turkmen vowels can be either long or short, except ‘e’ which must be short and ‘ä’ which must be long. Length is indicated by a colon in phonetic transcription, but is not indicated in Turkmen orthography. For example, the words for “grass” [ot] and “fire” [o:t] are both written ot, and “horse” [at] and “name” [a:t] are both at. In the table below, all long vowels have been indicated as such.

Form A	Form B	Translation
__(11)__	gaýtala:n	to repeat
__(12)__	daňan	to tie
__(13)__	hormatla:n	to respect
__(14)__	ýalka:n	to bless
__(15)__	bogan	to stifle
kowul	__(16)__	to expel
oýnal	__(17)__	to have fun
kakyl	__(18)__	to hit

(b) Fill in blanks 11-18, indicating length on all vowels.

△ ‘ö’ and ‘ü’ are pronounced like English long ‘a’ and ‘e’, but with rounded lips (like in German “zwölf”, “fünf”); ‘ä’ is like the ‘a’ in “cat”; ‘y’ is similar to English long ‘o’,

but without rounded lips; ‘š’ is like ‘sh’, and ‘č’ like ‘ch’, ‘ý’ is like English ‘y’, and ‘ň’ like ‘ng’ in “sing”. The remaining letters are pronounced similarly to English.

Problem 9. Nakanai (2020, NACLO, 1 pg)

Nakanai (Lakalai) is a language spoken by around 13,000 people of the Nakanai tribe in West New Britain, Papua New Guinea. Below are some words in Nakanai given along with their so-called “reduplicated form,” a grammatical form which has a large number of different uses (e.g. plural verb marking, habitative mood, and collective plurals). However, these uses are irrelevant to the derivation of the reduplicated form.

Underlying Form	Reduplicated Form
basi	baibasi
beta	babeta
galolo	galololo
kise	kekise
baimopu	baimoumopu
seku	seuseku
hugu	hugugu
sasa	sasasa
mari	marimari
toa	tatoa
kirosi	kirosirosi
ligi	ligiligi
kebo	kokebo
kevemuki	kevemuimuki
baharu	bahararu
golu	golugolu
mota	mamota
kedu	keikedu
vigilemuli	vigilemulimuli

- (a) Provide the reduplicated forms of the following words below **tahalo, sekela, pita, bake, loke, voro, valolohoka, pasi, kusa, bebe, hilo, sivo, rabu, tarile, sobe, vitaumetari, vituga**

Problem 10. Proto-Slavic (2022, Swedish LO, 2 pg)

Below are a number of words in the reconstructed language Proto-Slavic, their approximate meaning, and phonetic transcription of the words they gave rise to in Polish and Russian, but some words are missing.

Proto-Slavic	Polish	Russian	Meaning
lâsъ	las	l'ies	forest (NOM SG)
lâsæ	'lɛɛɛ	'l'ies'ie	forest (LOC SG)
môžъ	mow̃ż	muş	man (NOM SG)
môža	'mɛw̃ʒa	'muzə	man (GEN SG)
lědъ	lut	l'et	ice (NOM SG)
lědu	'lɔdu	__(1)__	ice (GEN SG)
mêso	'mjɛw̃sɔ	'm'asə	meat (NOM SG)
mêso	__(2)__	m'as	meat (GEN PL)
blōdъ	bwont	błut	invention (NOM SG)
blōda	__(3)__	'błudə	invention (GEN SG)
màsto	'mjastə	'm'iestə	place (NOM SG)
màstæ	'mjɛstɛɛ	'm'ies't'ie	place (LOC SG)
kvætъ	kfjat	tsv'iet	color (NOM SG)
kvætæ	'kfjɛtɛɛ	'tsviet'ie	color (LOC SG)
kòrl'ъ	krul	kɛ'rol'i	king (NOM SG)
korl'à	__(4)__	kəre'l'ja	king (GEN SG)
lādъ	wat	łat	order (NOM SG)
lādæ	'wad̃zɛ	'ład'ie	order (GEN SG)
golvā	'gwɔva	gəłɛ'va	head (NOM SG)
golvæ	'gwɔvjɛ	__(5)__	head (LOC SG)
rôbъ	rəmp	rup	navel (NOM SG)
rôbæ	'rɛmbjɛ	'rubə	navel (LOC SG)
bōrъ	bur	bor	conifer (NOM SG)
borù	__(6)__	bɛ'ru	conifer (LOC SG)
røkà	__(7)__	__(8)__	hand (NOM SG)
pōlzъ	__(9)__	'połəs	sledge (NOM SG)
pōlzæ	__(10)__	'połəz'ie	sledge (LOC SG)
dōrga	__(11)__	__(12)__	road (NOM SG)
__(13)__	__(14)__	słɔ'ga	pipe (NOM SG)
slùgъ	swuk	__(15)__	pipe (GEN PL)
__(16)__	'tʂaska	'tr'ieskə	cod (NOM SG)
NOT GIVEN	__(17)__	__(18)__	head (GEN PL)

(a) Fill in 1-18. For 16, 3 answers are possible, and for 18, 2 answers are possible. For these two, write all of the possible answers.

As Old Slavic is a reconstructed language, it is not known exactly how its sounds sounded, why the author has chosen to use an established phonemic transcription, with the exception that the sound usually written ě is here written æ to avoid stacked diacritics. In this transcription, ogonek (̃) is used for nasalized vowel, and circumflex

(ǃ), grave (ǂ) and double grave (ǃ) accent indicates different types of tone accent. The letters (ɿ) and (ʊ) stand for very short [i] and [u] sounds, ɿ' denotes the palatalized version of ɿ, and ž denotes a sound that was probably something like [ʒ]. Along with the approximate meanings of the words in this problem, case (NOM/GEN/LOC stands for nominative/genitive/locative) and number (SG/PL stands for singular/plural) for each word form.

Problem 11. Tocharian (2024, MSK)

Some verb forms of two closely related languages, Tocharian A and Tocharian B, and their English translations are given:

Tocharian A	English	Tocharian B	English
! pälskāt	he thought	! pälkānt	they saw
musnāl	he must climb	kotnatsi	to split
! skenāl	he must strive	paikāte	he wrote
kropnal	he must gather	käryātai	you (sg.) bought
! wenātrā	he drops	kraupātai	you (sg.) gathered
rinātrā	he desires	kārsnātrā	he knows
kālpāte	you (sg.) received	prutknātrā	he becomes full
wärpnāmtrā	we feel	mrausknātrā	he feels disgust for the world

- (a) Translate the following verb forms into English, indicating which language each belongs to. If multiple variants are possible, give them all: *kālpnāl*, *mrauskāte*, *wärpāte*, *pekat*, *kropat*, *musāt*, *käryāmte*
- (b) There are small differences between the different Tocharian A manuscripts. For example, in one of the manuscripts the dialectal form *kropnasi* was found. Indicate how this form should look in “standard” Tocharian A, and translate it into English.
- (c) One of the Tocharian A forms, marked with “!” in the table, is written with an error. Find and correct it.
- (d) Translate from English: *to know* (A), *he must feel* (A), *they drop* (A), *he felt disgust for the world* (A), *they feel* (A, B), *they collected* (A, B), *he split* (B).

⚠ Tocharian A and B are extinct Indo-European languages from the Tocharian branch, formerly spoken in the Tarim Basin.

Problem 12. Barbareño (2021, MSK)

The words of the Barbareño language are given in singular and plural forms, along with their English translations:

sg.	pl.	trans.
?onyoko	?on?onyoko?	shark
?ixpaniš	?ix?ixpa?niš	acorn
?awuxa	?aw?awuxa?	needle
yuxnuc	yuxyuxnuc?	hummingbird
tomol	tomto?mol	boat
takak	taktakak?	quail
taniw	tanta?niw	child
?elye?wun	?el?elye?wu?n	sword-fish
?amut'ey	?am?amut'e?y	sister
taxama	taxtaxa?ma	skunk
?išow	?iš?išo?w	coal
c'inowon	c'inc'ino?won	hill
xellex	xelxe?lex	hawk
puxu	?	dish
?ališaw	?	day
?uwu?mu	?	food
paxat	?	whale
?akayiš	?	bed
qowoč	?	salmon
kawayu	?	horse
q'emen	?	mouse
wulu?wul	?	lobster

(a) Fill in the plural forms.

⚠ Barbareño is an extinct Chumashan language formerly spoken on the central California coast.

Problem 13. Chakha (2024, MSK)

Some verbs of the Chakha language are given in the 3rd person singular masculine and in a special impersonal form, along with their English translations:

3 sg. m.	Impersonal	Translation
agbakābām	agwakāwim	to stuff
absabāsām	ab ^w šab ^w āšim	to go groping
barām	b ^w arim	to speak
g'ābārām	g'āb ^w ārim	to pay taxes
kābāsāsām	kāwāsāšim	to confuse
nākābām	nākawim	to find
tābāsām	tāb ^w āšim	to fry
ṭābanārām	ṭāwanārim	to roll up
abām	a_im	to give
abetātām	a_e_āčim	to expand
akābārām	akā_ārim	to celebrate
bəṭābātām	_ə_ā_ā_im	to stir
gərābātām	gərā_ā_im	to turn over
sānābām	sānā_im	to spin
tāzrabātām	_āzrab ^w āčim	to hope

(a) Fill in the blanks (marked _).

Problem 14. Mia (2022, MSK)

Below are some nouns of the Mia language in their singular and plural forms with English translations:

Sg.	Pl.	Translation
ábààngù	ábààngwángwàw	lizard
átár	átáràràw	squirrel
átlákwám	átlákwámámàw	spear
ávàkìy	ávàkìyáyàw	thorn/spike
caángú	caángwángwàw	stork
dèm	dèmámáw	tree
díngé	díngángáw	bench
dùwàkè	dùwàkákàw	horse
dzàbèrákú	dzàbèrákwákwàw	guinea fowl
gùmbè	gùmbàmbáw	calabash
kóm	kómámáw	hand
kèvèn	kèvènanàw	buffalo
mbərgu	mbərgwágwàw	ram
sày	sàyáyáw	liver
sértéhá	sértéháhàw	lake
ʔíjé	?	mortar
dèmbèr	?	baobab
káy	?	fight
kúnkúl	?	hat
kùtè	?	thing
zhááké	?	donkey
tàl	?	beer
tùwàtúw	?	body

(a) The author of the task accidentally forgot to indicate the tones in the singular form of the word ‘ram’. Do it for him.

(b) Fill in the blanks. Explain your solution.

△ Mia is a Chadic language spoken in northeastern Nigeria. ` , ´ above vowels denote low and high tones; the absence of a mark denotes a default tone level.

Problem 15. Georgian (2021, MSK)

Given (in Latin transcription) are some nouns and adjectives of Georgian and transitive verbs formed from them:

Source	Gloss	Verb	Translation
adamiani	person	gaadamianeba	to humanize
okro	gold	mookvra	to gild
bolo	end	daboloeba	to finish
giži	madman	gagižeba	to drive insane
brali	guilt	dabraleba	to blame
grili	cool (adj.)	gagrileba	to cool down
imedi	hope	daimedeba	to reassure
tuši	ink	moṭuṣva	to shade with ink
paṭenṭi	patent	daṭaṭenṭeba	to patent
ṣakari	sugar	moṣakvra	to sugarcoat / candy
qru	deaf	gaqrueba	to deafen
santeli	wax	mosantvla	to wax
ideali	ideal	gaidealeba	to idealize
vercxli	silver	movercxvla	to silver-plate
čala	straw	močalva	to thatch
žarimi	fine	dažarimeba	to fine
γobe	wattle	moγobva	to fence with wattle
čamali	medicine	?	to treat by spraying
avansi	advance	?	to advance (money)
cebo	glue	?	to coat with glue
suleli	fool	?	to fool
žildo	reward	?	to reward
kromi	chrome	?	to chrome
ṭudra	powder	?	to powder
uṇqlo	anhydrous	?	to dehydrate
muri	soot	?	to blacken with soot
picari	board	?	to board up

(a) Fill in the blanks.

Problem 16. Soqotri (2021, MSK)

The following are verbs of the Soqotri language, belonging to two verb classes (conventionally numbered I and II):

Class I	Meaning	Class II	Meaning
deme	to sleep	dami	to lull (to sleep)
ʔegod	to hit a target (of a thrown object)	ʔogid	to throw at a target
ʔeret	to inherit	ʔarit	to bequeath
feš	to be destroyed/scattered	fašiš	to destroy/scatter
bere	to give birth (to a child)	bari	to deliver a baby
leḳaṭ	to gather/pick (fruit)	loḳiṭ	to glean here and there
gedom	to cut off (e.g. a piece of meat)	godim	to butcher (a carcass)
ḳebaṣ	to fray (an edge)	ḳobiṣ	to hem
ṭeḳ	to be crumbled	ṭaḳiḳ	to crush/pound
ḳelob	?	ḳolib	?
deker	to remember	?	to make a kid bleat (to lure its mother)
gešol	to break	gošil	?
nemaṭ	?	nomiṭ	?
ṣeḳ	to bleat (of a kid)	?	to shatter (e.g. pots)
tebor	to beat	tobir	to cause someone to shed tears
beše	to cry	?	–

- (a) Determine which of the verbs **nemaṭ** and **nomiṭ** means ‘to be in good condition’, and which means ‘to comfort, to treat well’.
- (b) Determine which of the verbs **ḳelob** and **ḳolib** means ‘to turn over’, and which means ‘to twist (an object in the hands); to turn from side to side (of a bedridden patient)’.
- (c) Fill in the remaining blanks in the table.

Problem 17. Mbodomo (2020, MSK)

The roots and their past-tense forms of the Mbodomo language are given:

#	Root	Past	Translation
1	bérkìdî	bèrkàdà	to break
2	bídi	òdì	to separate
3	?	òtà	to peel (cassava)
4	?	bùfi	to scorch / singe
5	gbêŋ	?	to heat up
6	gbîn	gbìnì	to harvest
7	hêl	hèlà	to sob
8	kátitì	kàtàtà	to dry
9	kífìdì	kìfìdì	to answer
10	kímkìdî	?	to roll up
11	kóbìlì	?	to hang
12	kótì	kòtà	to pluck out
13	kúrsì	kùrsì	to clear / remove
14	?	kùtikì	to begin
15	?	mbèrà	to sing
16	mgbénzì	mgbènzà	to shake
17	gôm	?	to chop with a machete
18	wísì	wìsì	to hide
19	yámbìlì	yàmbàlà	to disappear
20	túkri	?	to turn

(a) Fill in the blanks.

△´ and ` above vowels mark high and low tones; ^ marks a falling tone.

Problem 18. Turkish (Stress) (2019, MSK)

Here are some Turkish words with stress marked:

#	Word	#	Word
1.	Ahmét	9.	Diyárbakır
2.	Ízmir	10.	Erzíncan
3.	günéş	11.	Beşíkdüzü
4.	Şaziyé	12.	uyurgezér
5.	Bakácak	13.	Yeniceóba
6.	Antálya	14.	Kahramánmaraş
7.	Túrkiye	15.	hünkarbeğendí
8.	Ramazán		

(a) Mark stresses on: koca, yumurta, Bursa, İstanbul, Ankara, Pamukova, Mustafa, Saruhanlı, Aksaray, İskenderun.

Problem 19. Kakabe (2019, MSK)

Here are some roots and their definite forms in Kakabe with the corresponding English translations:

Root	Definite	Translation
welo	weloe	bicycle
babu	baboe	camel
sunbo	sunbe	container
tarawo	tarawoe	road
waado	waadoe	umbrella
robu	robe	dress
mungu	munge	powder
jubaado	jubaadoe	hairstyle
bottu	botte	boot
sologo	sologe	fishing net
kiibo	kiiboe	dream
keemogo	keemoge	old man
kenbu	kenboe	coal
deemu	deemoe	chimpanzee
?	nōnbe	liana
doftooru	?	doctor
ganbo	?	pigeon
hodo	?	village
sengo	?	direction
programu	?	plan
dalilu	?	mind
luumo	?	market

(a) Fill in the blanks.

Problem 20. Khvarshin (2018, MSK)

Khvarshin verbs are given in the present and the so-called past I and past II tenses, along with their English translations:

Present	Past I	Past II	Translation
bužaha	bujža	bužana	to believe
lahekko	lahejk'a	lahek'na	to ignite / light
lahehe	lajla	lałena	to plow
lasso	lajsa	lasna	to take
liččo	lič'č'a	lič'na	to carve
piłaha	piłla	piłana	to blow
žikko	žik'k'a	žik'na	to beat
?	?	goł'ana	to call / summon
k'išehe	?	k'išena	to play
?	liqqa	liqena	to know
?	nit't'a	nit'na	to go out (extinguish)
?	?	tiłna	to give
žužulaha	?	?	to move

(a) Fill in the blanks.

Problem 21. Romani (2018, MSK)

Below are Romanian–Romani borrowings, with the English translations:

Rom.	Romani	Tr.	Rom.	Romani	Tr.
covrig	kovrigo	bagel	răcoare	řakar'a	coolness
delă	dela	case	putere	put'ar'a	power
treaz	tr'azo	sober	leagăn	l'agəno	cradle
roată	řata	wheel	scurt	skurto	short
rublă	řubla	ruble	vesel	v'aselo	cheerful
steag	st'ago	flag	poveste	pov'ast'a	story
toamnă	tamna	autumn	nepot	nepoto	grandson
pec	peko	tar	nepoată	nepata	granddaughter
vas	vaso	vessel			

- (a) Fill in the Romani forms: **moarte** (death), **păcală** (disgrace), **rătez** (hook), **neam** (relative), **vreme** (time).
- (b) Fill in the Romanian source forms given the Romani: **kr'anga** (branch), **nəkazo** (grief).

Problem 22. Hungarian (2018, MSK)

Geographical names in Hungarian are given in different case forms with their English translations:

Hungarian	English	Hungarian	English
Abonyba	in Abony	Mariborba	in Maribor
Amszterdamból	from Amsterdam	Miskolc	Miskolc
Csehország	Czech Republic	Miskolcra	to Miskolc
Esztergomba	in Esztergom	Miskolcról	from Miskolc
Hatvan	Hatvan	Nagybajomból	from Nagybajom
India	India	Nyíregyháza	Nyíregyháza
Indiába	to India	Ózd	Ózd
Kaposvár	to Kaposvár	Pomáz	Pomáz
Komárom	Komárom	Salgótarjánba	to Salgótarján
Lagos	Lagos	Sopronból	from Sopron
Oroszországból	from Russia	Tatabánya	Tatabánya
London	London	Tatabányára	to Tatabánya
Londonból	from London	Toronto	Toronto
Magyarországról	from Hungary	Torontóból	from Toronto

- (a) Translate into Hungarian: *in Lagos*, *to the Czech Republic*, *from Ózd*, *from Hatvan*, *in Pomáz*, *in Komárom*, *from Nyíregyháza*, *Salgótarján*.

Problem 23. Soqotri (2017, MSK)

Verbs of Soqotri are given in two past-tense forms (3sg.m. and 3sg.f.). Some vowels are missing (marked _):

3sg.m.	3sg.f.	Meaning
ʔedak	ʔidaʔo	to be heavy
ʔaraʃ	ʔeroʃo	to attack
ʃeber	ʃibaro	to see
ʃelaʃ	ʃilaʔo	to err
belaš	belašo	to be scarce
rekot	rekato	to trample
fezaʃ	fizoʃo	to fear
gamah	gemoho	to catch
herom	heramo	to pull out
zelaʔ	zelaʔo	to draw water
leseki	lisako	to stick
nafaʃ	nefoʃo	to do, to work
neker	nikaro	to long for home
ʃeʔaʃ	ʃiʔoʃo	to feel hunger
ʃebaʃ	ʃiboʃo	to be sated
terob	terabo	to go down
betaʔ	betaʔo	to stride
senem	sinamo	to rot
nekaʃ	niʔaʃo	to decrease
feraʃ	f_r_ʔo	to cut
sebah	s_b_ho	to be liquid
ʃ_f_ʔ	ʃ_f_ʔo	to wrap in cloth
d_l_k	d_l_ʔo	to be abundant
k_r_y	k_r_yo	to jump
s_b_h	seboho	to swim
ʔeʔah	ʔ_ʔ_ho	to feel pain
t_m_ʃ	t_m_ʃo	to urinate
f_ʔ_ʃ	f_ʔ_ʃo	to break

(a) Fill in the blanks.

Problem 24. Tibetan (2017, MSK)

There are four tones in Tibetan: level high (I), rising (II), rising-falling (III) and falling (IV). Some Tibetan words are given (in Latin transcription) with the tone with which they are pronounced:

Word	Gloss	Tone	Word	Gloss	Tone
dku	side	I	ston	autumn	I
kun	all	I	do	pair	II
dpañs	height	IV	sde	tribe	II
mgo	head	II	ldebs	surface	III
nub	west	III	mnog	profit	IV
gdugs	umbrella	III	kug	hook	IV
sad	frost	IV	dnul	silver	I
re	every	II	gañs	snow	III
su	who	I	lkog	secret	IV
til	sesame	I	bod	Tibet	III
gzig	leopard	III	blo	mind	I
mñan	boat	I	pag	brick	IV
ldum	vegetables	II	ñan	character	II
glag	eagle	IV	zil	luster	II
sñigs	sediment	IV	ñams	feeling	III

- (a) Determine which tone the following words have: **mi** ‘person’, **dmañs** ‘people’, **kloñ** ‘wave’, **sbugs** ‘cavity’, **bur** ‘bolt, latch’, **mdog** ‘color’, **lug** ‘sheep’.

Problem 25. Irish (2017, MSK)

Below are Southern-Irish words with the stress on the long vowel shown by the acute accent, and their English glosses:

Irish	Gloss	Irish	Gloss
agallamh	conversation	codlatán	sleepyhead
cúramach	careful	dílárú	decentralization
fágálach	idler	tobathrú	sharp change
ísiltír	lowland	úllóga	apple pies
minice	frequency	órshúlach	syrup
mónadán	cranberry		

- (a) One of the following words has an error in the stress. Correct it: **ullmhúchán** ‘preparation’, **gramadach** ‘grammar’, **údarás** ‘power’.
- (b) Determine and mark the place of stress in the following words:

Word	Gloss	Word	Gloss
acadamh	academy	grástúlacht	elegance
amadán	fool	ochtódú	eightieth
amárach	tomorrow	ózónú	ozonation
lúnadán	pimple	scáthanna	shadows

Problem 26. Catalan (2016, MSK)

The Catalan language originated from Latin. Latin and corresponding Catalan words are given, along with English translations. Latin nouns and adjectives are in the ablative, verbs in the infinitive:

Latin	Catalan	Translation
aprīle	abril	April
cadūcō	caduc	about to fall
sēculō	segle	century
arbore	arbre	tree
bibere	beure	to drink
alterō	altre	other
rōbore	roure	oak
ardōre	ardor	heat
circulō	cercle	circle
lātrāre	lladrar	to bark
pulsāre	polsar	to press
pipere	pebre	pepper
līberō	lliure	free
duplāre	doblar	to double
asperō	aspre	rough
comite	comte	count
computō	?	to calculate; account
lepore	?	hare
rūmōre	?	rumor
masculō	?	masculine
populō	?	people; nation
dēbitō	?	debt
fundere	?	to melt
sacrātō	?	sacred
socerō	?	father-in-law

(a) Fill in the blanks.

(b) Try to determine what the Latin source of Catalan *finir* ‘to end’ looked like.

Problem 27. Attié (2016, MSK)

The following are nouns of the Attié language in indefinite and definite forms:

Indefinite	Definite	Translation
kĩ	kĩĩ	clay jug
nõ	nõẽ	trap
be	beo	bird
dɔ	dɔɛ	hour
mũdũ	mũdũũ	snail
tʃĩ	tʃĩĩ	monkey
te	tee	horn
ko	koo	one of several wives
pẽ	pẽĩ	wild boar
sũ	sũĩ	house
bo	boe	stone
ji	jiu	chief
tɛ	tɛɔ	porcupine
atubu	atubui	taro (root)

- (a) Form the definite of each: **kokwĩ** ‘mouse’, **fe** ‘bone’, **gbě** ‘stick’, **alebo** ‘crocodile’, **tsu** ‘elephant’, **dzo** ‘mushroom’.

⚠ Attié is a Kwa language spoken in Côte d’Ivoire. ~ marks nasalisation.

Problem 28. Yaka (2015, MSK)

The verbs of Yaka are given in their imperfective and perfective aspect forms, with English translations:

Form 1	Form 2	Translation
fufuka	fufukidi	to crumple
kota	kotele	to enter
kana	kanini	to decide
sala	sadidi	to do
longa	longele	to teach
sweeka	sweekele	to hide
bila	bididi	to boil
kaanda	kaandidi	to glue
hita	hitidi	to precede
toma	tomene	to improve
bema	bemene	to glow red-hot
koola	koolele	to whistle
diimbuka	diimbukidi	to omit / let pass
nuunga	nuungini	to defeat
miituka	miitukini	to get angry
?	nokene	to drip
mina	?	to swallow
saamba	?	to speak in court
meeka	?	to bleat
vota	?	to vote
lela	?	to lull
niingika	?	to be intended for

- (a) Fill in the blanks.

- (b) Form 2 of the verb ‘to accuse’ is **ludidi**. Determine Form 1 and explain why this is the only possibility.

Problem 29. Adyghe (2015, MSK)

The verbs of Adyghe are given in the 3sg. present and future forms (in Latin transcription) with English translations. Some forms are omitted:

Present	Future	Translation
jewe	jeweš't	(he) will hit
jeλ.ə	əλ.əš't	(he) will strangle
ješxə	əšxəš't	(he) will eat
jewəce?wə	əwəce?wəš't	(he) will chew up
jefəte	əfəteš't	(he) will peck off
jedəsə	jedəsəš't	(he) will push
jecaqe	jeceqeš't	(he) will bite
jebzejə	jebzejəš't	(he) will lick
jeq ^w əte	əq ^w əteš't	(he) will break
jewəλeg ^w ə	əwəλeg ^w əš't	(he) will trample
jewəp·č'ate	əwəp·č'eteš't	(he) will chop down
ječ'əbawe	ječ'əbeweš't	(he) will pat (on the back)
jebewə	jebewəš't	(he) will kiss
jedə	ədəš't	(he) will sew up
?	əxəxəš't	(he) will swallow
jewə?wə	?	(he) will peck
jetxə	?	(he) will write
?	əwəbeteš't	(he) will smash
je?e	?	(he) will touch
jewəč'ə	?	(he) will kill
jepšeda?e	?	(he) will grab by the scruff
je?ex ^w ə	?	(he) will scratch
jebzə	?	(he) will cut

(a) Fill in the blanks.

Problem 30. Tundra Nenets (2016, RoLO)

Here are some phrases in Tundra Nenets and their English translations in random order:

- | | |
|--------------------|--------------|
| 1. yal'a | A. autumn |
| 2. xayer | B. dawn |
| 3. xa'masi | C. day |
| 4. yal'a tarp | D. fire |
| 5. xə' | E. lightning |
| 6. sar'olank yal'a | F. raindrop |
| 7. sar'o səwkocya | G. rainy day |
| 8. wəba' xa'm | H. storm |
| 9. xəxə' sar'o | I. sun |
| 10. xəxə' ty' | J. thunder |
| 11. tarpa'si | K. to appear |
| 12. ty' | L. to fall |
| 13. sar'omsi | M. to rain |

(a) Determine the correct correspondences.

(b) Translate into English: *səwkocya, sar'o*.

(c) Knowing that *pii'* means *night*, translate into Tundra Nenets: *dusk, leaf, stormy night, sunny day*.

Problem 31. (2018, OLM)

Here are some phrases in Khmer and their English translations in random order:

1. <i>hmpúur mət</i>	A. <i>anus</i>
2. <i>hmpúur tñəh</i>	B. <i>pinna (earlobe)</i>
3. <i>hntú m̀h</i>	C. <i>eyelid</i>
4. <i>hntú tñəh</i>	D. <i>snot (nasal mucous)</i>
5. <i>hntú ʔiak</i>	E. <i>adam's apple</i>
6. <i>káap kapoʔ</i>	F. <i>cavity (tooth decay)</i>
7. <i>khúul káap</i>	G. <i>lip</i>
8. <i>khúul mət</i>	H. <i>nipple</i>
9. <i>kləəŋ kók</i>	I. <i>eyeball</i>
10. <i>kləəŋ mət</i>	J. <i>milk</i>
11. <i>lá hrmèey</i>	K. <i>beard</i>
12. <i>òm mət</i>	L. <i>nose</i>
13. <i>òm m̀h</i>	M. <i>eyelash</i>
14. <i>òm p̀</i>	N. <i>tear</i>
15. <i>prlè p̀</i>	O. <i>mouth</i>
16. <i>ʔiak ráaŋ</i>	P. <i>chin/lower jaw</i>

(a) Determine the correct correspondences.

(b) Translate into English: *hntú hrmèey, ráaŋ òm p̀*.

(c) Translate into Khmer: *cave, stone, skin*.

Problem 32. Hmong (2021, UKLO)

Here are some phrases in Hmong and their English translations in random order:

- | | |
|-------------------------------|-----------------------------------|
| 1. daim nplooj | A. <i>the book</i> |
| 2. daim ntawv | B. <i>the breast</i> |
| 3. kua mis | C. <i>the coin</i> |
| 4. kua txiv hmab | D. <i>the fruit</i> |
| 5. lub mis | E. <i>the grape juice</i> |
| 6. lub ntaws | F. <i>the leaf</i> |
| 7. lub nyiaj | G. <i>the message (spoken)</i> |
| 8. lub txiv | H. <i>the message (written)</i> |
| 9. phau khaub | I. <i>the milk</i> |
| 10. phau ntawv | J. <i>the navel</i> |
| 11. phau nyiaj | K. <i>the nerves</i> |
| 12. tsab ntawv | L. <i>the pile of clothes</i> |
| 13. tsab xov | M. <i>the postal letter</i> |
| 14. txoj hmab | N. <i>the road</i> |
| 15. txoj kev | O. <i>the sheet of paper</i> |
| 16. txoj kev thaj yeeb | P. <i>the stack of bank notes</i> |
| 17. txoj leeg | Q. <i>the umbilical cord</i> |
| 18. txoj ntaws | R. <i>the vine</i> |
| 19. zaj xov | S. <i>the way of peace</i> |

(a) Determine the correct correspondences.

(b) Fill in the blanks:

1. __ (1) __ **hnub** *the sun*
2. __ (2) __ **hlua** *the rope*
3. __ (3) __ **klauj** *the song*
4. **sau** __ (4) __ *the essay*
5. __ (5) __ *the field*
6. __ (6) __ *the grape*

⚠ In the standard orthography used here, a syllable-final consonant letter indicates a consonant sound if it is **m**, but otherwise it indicates a tone; for example, the **-s** in **mis** indicates low tone (so, **mis** = /mì/).

Problem 33. Agbirigba (2021, IOL)

Agbirigba is a cant, or secret language, used by around 30 speakers of the Ogbakiri dialect of the Ikwerre language. Here are some words in the Ogbakiri dialect, along with their Agbirigba forms and translations:

Ogbakiri	Agbirigba		Ogbakiri	Agbirigba	
ńdá	ńtódá	father	rímé	tírítúmé	inside
bísí	túbítísí	poison	mónô	tómótônô	oil
àyò	òtìyò	amicable gesture	ìbèrè	ítùbètìrè	belch
ńbórô	ńtóbótórô	dream	áyâ	átíyâ	eye
ótí	ótítí	leg	ónû	ótônû	mouth
òtú	òtútú	to look for	móyô	tómótíyô	urine
míní	tímítíní	water	ngàdã	ntògàtòdã	chair
wèrù	tùwètùrù	to have	ómìrìmà	ótòmìtìrìtòmà	meat, flesh
èbó	ètóbó	two	àgìrìgò	àtìgìtìrìtògò	defiance
àwâ	àtówà	to break	wéné	túwétíné	siblinghood

(a) One of the Agbirigba words above is an exception. Its form is not determined by the same rules which determine the rest of the Agbirigba words. Which is it? What would its form be if it were determined by the same rules?

(b) Give the Agbirigba forms of the following Ogbakiri words:

1. rótô chewing-stick
2. rèwó deity
3. bíkó please
4. jýé rìyà female
5. békʷó fire
6. ríyòrò charcoal
7. hʷəgà go out!

△ Ikwerre belongs to the Igboïd group of the Atlantic-Congo family. It is spoken by approx. 200,000 people in Rivers State, Nigeria.

The words are given in a simplified transcription. The mark ~ indicates the nasal pronunciation of the vowel. The marks ´, ` and ^ indicate high, low, and falling tone, respectively. ŋ = ng in hang; ɲ ≈ ny in canyon; y = y in yay!; tʃ = ch in church. kʷ and hʷ are pronounced as k and h respectively, but with the lips rounded. ε, ɪ, ɔ and ʊ are pronounced as e, i, o and u respectively, but with the root of the tongue further back in the mouth.

Problem 34. Scottish Gaelic (2022, ČLO)

Here are some phrases in Scottish Gaelic and their English translations in random order:

1. comhachag-bhàn	A. stormy sea
2. comhachag-bheag	B. zucchini
3. inneal-nighe	C. pumpkin
4. mathan-bàn	D. silk
5. mathan-molach	E. polar bear
6. mathan-nighe	F. grizzly bear
7. mearag	G. raccoon
8. mearag-bheag	H. washing machine
9. muir-molach	I. laundry room
10. sìoda	J. velvet
11. sìoda-molach	K. barn owl
12. taigh-beag	L. common owl
13. taigh-nighe	M. toilet

(a) Determine the correct correspondences.

(b) Knowing that:

bòrd-cofaidh	coffee table
biadh-sòghail	luxury food
ad-chruaidh	protective helmet
clàr-cruaidh	hard disk
uisge-talmhainn	ground water
cnuimh-thalmhainn	earthworm

Translate into Scottish Gaelic:

1. white house
2. coffee maker
3. silkworm larva
4. furry cap
5. snack

Notes:

1. Velvet: A soft fabric, from the base layer of which dense short threads emerge. It was traditionally made of silk.
2. Barn owl: A medium-sized owl, one of the most common owls. It features a white heart-shaped "face".
3. Common owl: A small owl, about a quarter of a meter high. It has distinctive yellow eyes and a white stripe above the eyes.

4. The silkworm is a nocturnal butterfly whose larvae form the fiber from which silk is made.

Problem 35. Dioula and Mossi (2022, OLR)

Here are some words in Dioula and their English translations in random order:

1. sotigi	A. grandfather
2. denmuso	B. professor (f)
3. tigi	C. head of the family
4. wolo	D. biological mother
5. woloba	E. oldest brother
6. kòròkè	F. old woman
7. mòkè	G. man
8. mòmuso	H. girl
9. kàlànsò	I. property
10. falatò	J. grandmother
11. adàmàden	K. mother
12. ba	L. woman
13. mùso	M. school
14. mùsòkòròba	N. to give birth
15. kàlàmba	O. orphan

(a) Determine the correct correspondences.

(b) Translate into English: mòden, kàlàmba, kàlàmben.

Here are some more words and phrases in Mossi and their English translations in random order:

16. bug-těnga	P. honesty
17. ma	Q. hell
18. těng-n-biilim	R. nationality
19. bug-koom	S. ground floor
20. gūbgi	T. student
21. těngr roogo	U. island
22. ko-solsga roogo	V. bathroom
23. ma-biilim	W. mother
24. burkĩndlim	X. kerosene
25. ko-gūb-těnga	Y. citizen
26. karen-biiga	Z. brotherhood
27. těng-n-biiga	AA. to surround

- (c) Determine the correct correspondences.
- (d) Translate into English: **koom, bugum, karen-doogo**.
- (e) What is the literal meaning of the country name *Burkina Faso*?

⚠ Dioula and Mossi belong to the Niger-Congo language family. It is spoken by 7 and 8 million people in West Africa respectively. Both languages are the official languages of Burkina Faso.

Problem 36. Manx (2017, MSK)

The following Manx phrases are given with their English translations in random order:

- | | |
|----------------------------|---------------------------|
| 1. Jerry Geuree | A. <i>June</i> |
| 2. mean oie | B. <i>January</i> |
| 3. Toshiaght Souree | C. <i>midnight</i> |
| 4. Oie gyn cadley | D. <i>February</i> |
| 5. Jerry Souree | E. <i>July</i> |
| 6. cadley geuree | F. <i>hibernation</i> |
| 7. Toshiaght Arree | G. <i>May</i> |
| 8. Mean Souree | H. <i>sleepless night</i> |

- (a) Determine the correct correspondences.
- (b) Translate into English: **Mean Fouyir, gyn jerry**.
- (c) Translate into Manx: *April, October*.

⚠ Manx is a Celtic language spoken by about 1,800 people on the Isle of Man.

Problem 37. Georgian (2015, MSK)

English phrases and their Georgian translations (in Latin transcription) are given in random order. Some English phrases correspond to two Georgian translations:

English: without tasty buns, with cinnamon, without ice cream, with this butter, without hazelnuts, with almonds, with tasty cream, without almond butter, with hazelnut seasoning

Georgian: gemrieli naḡebit, am zetit, nuṡiani, txilis saneleblit, gemrieli puntuṡis gareṡe, nuṡit, daričiniāni, utxilo, naḡinis gareṡe, daričinit, nuṡis zetis gareṡe, unaḡino, txilis gareṡe

- (a) Determine the correct correspondences.
- (b) Write in Georgian in all possible ways: *with cream, with creamy ice cream, without almonds, without this seasoning*.

Problem 38. Afrihili (2020, MSK)

The following are nouns from the constructed Afrihilian language with their English translations:

Afrihili	English	Afrihili	English
adu	tooth	ikalini	pens
ajamuri	republic	ijamura	president
akalini	pen	ilengi	horses
afimadi	driver	imukazi	girls
amkate	bread	isabamatu	cobbler
amola	children	ishule	school
amukamo	kingdom	omuntu	person
aturesine	bouquet	olengi	horse
afidi	automobile	uruzi	river
emeli	ship	omola	child
uruzindi	stream	omukazi	girl
eshuli	school principal	emelisini	fleet
eture	flowers	oluganda	dialect
achosino	school of fish	omuntundu	dwarf

(a) Fill in the blanks below:

dentist	?	group of girls	?
ships	?	fish	?
?	ajamura	?	oluga
flower	?	shoe	?
bread	amkamate	king	?

(b) For each Afrihili word, indicate the most likely translation: **A. imulenzi:** (a) fruit; (b) boys; (c) bridge. **B. aposino:** (a) luggage; (b) classroom; (c) relative. **C. iwelemase:** (a) book; (b) library; (c) librarian.

△ Afrihili is a constructed pan-African auxiliary language created by Kumi Attobrah in 1970.

Problem 39. Lithuanian (UKLO, 2018)

Four Lithuanian friends, Danute (f), Jokubas (m), Matis (m) and Regina (f) are planning a trip. Here are some extracts from their conversation. Pay attention to the diacritics

Speaker	Lithuanian	Translation
Danute to Jokubas	Mes n�rime gra�os.	We want some change.
Jokubas to Danute	A� n�riu �em�lap�.	I want a map.
Danute to Regina	Jis ska�ito.	He's reading.
Jokubas to Matis	Ar tu turi gra�os?	Do you have some change?
Matis to Jokubas	A� neturiu gra�os.	I don't have any change.
Matis to the others	J�s einate.	You're going.
Regina to the others	Mes turime gra�os.	We have some change.
Matis to Regina	Tu turi dvirat�.	You have a bike.
Matis to the others	Regina turi dvirat�.	Regina has a bike.
Jokubas to Danute	Ska�tyk �em�lap�.	Read the map!
Danute to Matis and Regina	J�s neska�tote �em�lap�.	You aren't reading the map.
Matis and Regina to Danute	Mes einame.	We're going.

(a) From the conversation, work out how to translate these English sentences into Lithuanian.

Speaker	English
to Danute and Jokubas	Do you have a bike?
to Jokubas and Matis	You're not reading.
to Matis	You are going.
-	Is Matis going?
to Matis	We don't want any change.
Jokubas to the others	You don't have any change.
-	Don't I have a bike?
-	I don't want a map.

  Lithuanian is the official language of Lithuania, and is one of the surviving languages in the Baltic branch of the Indo-European family. As a result of its isolation, Lithuanian has retained many characteristics of the Indo-European ancestral language, making it particularly interesting for linguists.

Problem 40. Ainu (HKLO, 2020)

Given below are some sentences in the Shizunai dialect of Ainu¹ and their translations into English.

Ainu	English Translation
korpa as wa isam	<i>We have had.</i>
inkartek an wa an	<i>I was glancing.</i>
e inkar wa an	<i>You (sg) were seeing.</i>
inu wa isam	<i>He has listened.</i>
iperepa wa oka	<i>They were feeding..</i>
e ipe wa an	<i>You (sg) were eating.</i>
eci inuruypa wa oka	<i>You (pl) were listening a lot.</i>
cie korettek wa isam	<i>We have lent you (sg).</i>
cieci nukarruypa wa isam	<i>We have stared at you (pl).</i>
eun nurepa wa oka	<i>You (sg) were telling us.</i>
un etekpa wa oka	<i>He was tasting us.</i>
ecien nutek wa an	<i>You (pl) were listening to me a little.</i>
an yaynu wa isam	<i>I have thought.</i>
an eruypa wa oka	<i>I was devouring them.</i>
inuruypa as wa isam	<i>We have listened a lot.</i>
en e wa an	<i>They were eating me.</i>
e yaykore wa isam.	<i>You (sg) have given yourself (sg).</i>
cieci nurepa wa oka.	<i>We were telling you (pl).</i>

- (a) Translate into English. If you think that there are multiple possible translations, write them all:

e nukarepa wa isam e koreruy wa an
ci yaynukarpa wa oka nuruypa wa isam

- (b) Translate into Ainu:

He was listening to you (pl). We have eaten.
You (sg) were thinking a lot. They were staring.
We were borrowing him. I have glanced at them.

¹ Ainu is an indigenous language of Japan that is unrelated to Japanese. It is now almost extinct, but was previously spoken by the Ainu people around the northern island of Hokkaido up until the last century.

Problem 41. Gilbertese (NACLO, 2018)

The Gilbertese language is spoken by most of the over 100,000 people who live in the Pacific island nation of Kiribati. Here are some sentences in Gilbertese with their English translations:

Gilbertese

ko nakonako ŋkoe

e nakonako te aiine

i takaakaro ŋai

e nakonako nakon te titooa Meeri

a tekateka irarikin taian auti taian aiine

a tebotebo taian nakekei n te bong aei

i tebotebo inanon te auti ŋai

a takaakaro inanon te titooa taian ataei

ko tekateka ŋkoe ningaabong

a takaakaro irarikin te kawai taian ataei n te bong aei

Translation

You are walking.

A woman is walking

I am playing.

Mary is walking to the store.

The women are sitting next to the houses.

The people are bathing today.

I am bathing in the house.

The children are playing in the store.

You will sit tomorrow

The children are playing next to the road today.

(a) Below are some Gilbertese sentences with the word order scrambled. Write the sentence with the correct word order and then translate into English.

1. bong n Meeri auti e nakon te te nakonako aei

2. inanon ŋai kawai i te tekateka

3. ataei a nakon taian ningaabong kawai nakonako te

(b) Translate into Gilbertese:

4. The women will play tomorrow.

5. You are sitting next to the store today.

Problem 42. Yanyuwa (IOL, 2024)

In Yanyuwa, men and women have some differences in how they speak. Here are some sentences in the women's dialect of Yanyuwa and their English translations:

1. **nya-ardu kiwa-wani**

The boy returned

2. **rru-bardibardilu kanda-wubala ma-burlurlu**

The old woman will bake the bread.

3. **nyu-marralngujilu jarrilu-yabimanji rra-walkuru**

The doctor is healing the pregnant woman.

4. **wakuku ji-wulumanji**

The dog is running.

5. **linji-nhanawalu kanyalu-wudurrumala nya-mirningiya**

The women will feed the man.

6. **ji-jakudukudulu kanyinju-athama nya-munanga**

The emu chased the white man.

7. **nyu-ardulu jilu-nganji warrbima**

The boy is looking at the dingo.

For contrast, below are some sentences in the men's dialect and their English translations:

8. **li-walkuru kalu-walanyma**

The pregnant women emerged.

9. **ki-warrbimalu kilu-tha wunala**

The dingo ate the kangaroo.

10. **rra-bardibardi janda-waninji**

The old woman is returning.

11. **ki-bulijimanjilu karrilu-yngkarri rra-ardu**

The policeman heard the girl.

12. **ki-mirningiyalu jilu-wubanji ma-ngarra**

The man is cooking the food

13. **rru-yanyuwalu kanda-ngala munanga**

The Yanyuwa woman will look at the white man.

14. **linji-marralngujilu kalalu-yabimala li-malbu**

The doctors will heal the old men.

(a) Translate into English:

15. **mirningiya ka-wulumala**

16. **nyu-munangalu kalilu-wudurruma li-wakuku**

17. **linji-ardulu jalu-wubanji ma-burlurlu**

18. **rru-nhanawalu janda-athamanji ardu**

(b) For each of the Yanyuwa sentences in (a), indicate whether it is in the women's dialect, the men's dialect, or whether it is unclear.

(c) Translate into Yanyuwa, giving both men's and women's options if these differ:

19. *The dog chased the girl.*

20. *The pregnant woman is feeding the old women.*

21. *The old man healed the Yanyuwa man.*

22. *The kangaroo will return.*

⚠ The Yanyuwa language belongs to the Ngarna branch of the Pama–Nyungan family. It is currently spoken fluently by three women living in the Northern Territory, Australia. $k \approx g$ in lag. $nh = n$ in month. $rr = rr$ in Spanish perro. $th = th$ in leather. r , rd , rl and rn are pronounced like rr , d , l and n but with the tip of the tongue curled

backwards. n before j and d indicates some air flows through the nose before the consonant. Dingos and emus are animals found in Australia.

Problem 43. Haya (ONL- Romania, 2022)

Here are some sentences in Haya, written in the International Phonetic Alphabet, and their corresponding English translation:

Haya	English Translation
nogj e:mikono	<i>I'm washing my hands.</i>
end̥ʒw e:zo bikamubonwa	<i>Those houses were seen by him/her</i>
omwa:n a:kakuhend o:mukon o:gwo	<i>The child broke that hand of yours.</i>
ŋkutʃumbil e:nfulu ibili	<i>I am cooking the two fish for you</i>
bilibon e:ŋkoko ifatu	<i>They will see the three chickens</i>
ahend e:nd̥ʒu	<i>He/She breaks the house/houses</i>
eŋkok a:litʃumbwa	<i>The chicken will be cooked</i>
omwa:n o:gw a:kaj̥p̥enda	<i>That child hugged me</i>
olimutobolil e:ŋkok e:gjo	<i>You will pierce that chicken for him/her</i>
nlibitobola	<i>I will pierce them</i>
enfulw a:kambon e:mikon e:bili	<i>The fish saw my two hands</i>
aba:n a:bo bje:p̥enda	<i>Those children are hugging each other</i>
ako:gj o:mwa:na	<i>He/She washed the child</i>

(a) Translate into English:

netobola	ako:gj e:nfulu
ŋpend e:nd̥ʒw e:zo	olimbon o:mukon o:gwo
eŋkoko bilikutʃumbilwa	akahendil a:ba:na baʃatw a:b e:nd̥ʒu ibili

(b) Translate into Haya:

- You will be cooked.
- He/She hugs me.
- That house is pierced.
- He/She washed the three chickens for me.
- I saw their hands.
- Those two children will cook the fish for themselves.

△ The haya language is a member of the bantu language family. It is spoken by approximately 1.3 million people in Tanzania. ŋ = ng in 'king', ɲ = ny in 'canyon', ʃ = sh in 'ship', d̥ʒ = j in 'jump', tʃ = tch in 'Tchaikovsky'. 'ː' after a vowel indicates that the vowel is lengthened

Problem 44. Mee (APLO, 2019)

Here are some sentences in Mee and their possible English translations:

Mee	English Translation
yanameege	<i>You(sg) brought me for him today.</i>
nakadoogi	<i>He saw you(sg) for me today.</i>
teenedoogi	<i>He did not see us today.</i>
niyanedoope	<i>You(sg) saw us for us</i>
ebukaipa	<i>I saved them.</i>
meege	<i>You(sg) came today.</i>
kaaaakadoope	<i>We two met for you(sg).</i>
naakawageegai	<i>They fought for me today.</i>
aanabukaipaa	<i>You two saved me.</i>
teekameege	<i>We did not bring you(sg) today.</i>
teedokaipa	<i>You(sg) did not carry him.</i>
teekiyaaameepe	<i>We two did not come for you(pl).</i>

(a) Translate into English:

aameepai eekedoogi
naameegaa teenadokaipaa

(b) Translate into Mee:

- You(sg) saw them today.
- I did not come for him today.
- You two saw him for us.
- They saved each other.

(c) These sentences can be interpreted in different ways. Give all possible translations.

kameepi edoope

△ Mee (Ekari) belongs to the Trans–New Guinean family. It is spoken by approx. 100,000 people in Papua province, Indonesia.

Problem 45. Ubykh (2024, MSK)

Some Ubykh verb forms are given with their English translations:

Ubykh	English
səq ^w et ^w ən	I stop
ŝ ^w əq ^w eχan	You(pl) stop
adəχan	They rise
wəsəq ^w et ^w ən	I stop you
aš ^w əq ^w et ^w ən	We stop him
ŝ ^w əzχedəχan	I lift you (pl.)
səwdəq ^w et ^w ən	You stop me
awχeq ^w eχan	You stop them
wədəpət ^w ən	He lifts you
sadəq ^w et ^w ən	They stop me

(a) Translate into English: **adət^wən, wəš^wədət^wən, awdəq^wet^wən, až^wχeq^weχan, š^wəχedəχan.**

(b) Translate into Ubykh: *You get up., I lift him., We stop them., You(pl) lift us., He stops you.*

△ Ubykh is an extinct Northwest Caucasian language; the last native speaker died in 1992.

Problem 46. Abaza (2018, MSK)

Below are some Abaza verb forms with their English translations:

Abaza	English	Abaza	English
xIɣaɪɪd	we go here	ɕɣaɪɪd	?
ɕɣnaɕɪɫɪuam	I do not enter there	dɣɪmɕɪɕɪd	?
dɣɪakɪɪd	he came out here	ɕɣɪχɪuam	?
dɣɪaɪɕɪɪd	he watches here	dχɪɪd	?
ɕɣɪmχɪɪd	I did not calm down	xIɣɪnaɪɪuam	?
ɕɣɪnaɪɕɪɪd	I did not look there	?	he enters there
xIɣɪɣɪamɕɪɪd	we did not enter here	?	he did not come out here
dɕɪɪɪd	he shouted	?	he does not shout
ɕdɕɪɪd	I dig	?	we dug
xIɣɪɕɪuam	we do not watch	?	I did not watch here
		?	he did not dig

(a) Fill in the blanks.

Problem 47. Abaza (2020, MSK)

The verb forms of Abaza and their English translations are given:

#	Abaza	English
1	йббада	Whom did you see?
2	йылхIвайа	What did she say?
3	йбцхърагIайа	What helped you?
4	бызрыхIада	Who frightened you?
5	йылрыхIада	Whom did she frighten?
6	йылзрыхвгIада	Who sold that to her?
7	йызбрыхвгIада	To whom did you sell that?
8	дызцхърагIада	Whom did she help?
9	йылщцада	Who became accustomed to her?
10	бызпшыа	What did you look at?
11	бабабзаа	Where did you live?
12	йабалба	Where did she see that?
13	банбалцхърагIа	When did you help her?
14	данбагIай	When did she arrive?

- (a) Translate into English: йгIайда, данбабзаа, дызпшыа, дызрыхIайа, йылбрыхвгIайа, йызцхърагIайа.
- (b) Translate into Abaza: *Who helped her?, What did you say?, When did she scare you?, Who looked at you?, What did you get used to?*

Problem 48. Abaza (2023, MSK)

Given (in Latin transcription) are some Abaza verbs and their English translations:

- **crapsara** — to throw after
- **čəɫ^wcarə** — to drive out of the hearth
- **čəɫ^wpsara** — to throw out of the hearth
- **g^waragara** — to bring into the yard
- **g^warq'ara** — to jump out of the yard
- **bəž'pš'ra** — to peek out of a gap
- **bž'aɫ^wəzra** — to slip into a gap

- (a) Which of the Abaza verbs **napəçzara** and **q^wdaɬlara** is translated as 'to snatch from hands', and which as 'to fall on the neck'?
- (b) Translate into English: **g^wargara**, **čɫ^waplara**, **bž'apš'ra**, **čəɫ^wq'ara**, **napəçɫ^wəzra**.
- (c) Translate into Abaza: *to tear off from the neck, to drive into a gap, to throw on the neck, to tear off from behind.*

Problem 49. Dorumi-Koki (2023, MSK)

Below are some sentences in Dorumi-Koki and their English translations:

- | | |
|--------------------------------------|---|
| 1. Na vareyakari ya dairiyo. | While I was sleeping, you _{sg} returned. |
| 2. Na nega ago niyaka. | While I was descending, I spoke. |
| 3. Una nega ita ruyafa. | While we were descending, we gathered brushwood. |
| 4. Ya iruku iriyoma ina dairiyadi. | When you _{sg} had eaten, they returned. |
| 5. Ya ita rusi koru iriyadi. | When you _{pl} had gathered the brushwood, you _{pl} drank water. |
| 6. Ina tauga iriyori una ago niyafa. | While he was eating a banana, we talked. |
| 7. Ina neyoma na boyaka. | When he came down, I went up. |
| 8. Ina dairisi iruku iriyadi. | When they returned, they ate. |

(a) Translate into English: *Ina boga koru iriyadi., Ya varesi boyo., Na dairiyakama ina tauga iriyo., Ya ago niyadiri na iruku iriyaka.*

(b) Translate into Dorumi-Koki: *While we were sleeping, you (pl.) ate., When you (sg.) drank water, you (sg.) descended., When you (pl.) returned, I slept., While he was talking, he gathered brushwood.*

(c) Try to translate the words *iruku* and *ago* into English.

Problem 50. Turkish (2023, MSK)

Here are Turkish words with stress marked, and their English translations in random order:

Turkish: çöméziz, heykél, heykelimíz, şahínim, seçmenín, seçménsiniz, tacirín, tacíriz, yaverím

English: you are voters, my assistant, we are newcomers, we are traders, our statue, statue, your voter, your trader, I am a hawk

(a) Determine the correct correspondences. Briefly explain your solution.

(b) Translate into Turkish (with stress marked): *merchant*.

(c) Marking the stress, translate into English: *çömezsın, seçmenim, şahininiz*.

Problem 51. Soqotri (2022, MSK)

Below are some sentences in Soqotri and their English translations:

- | | |
|------------------------------|----------------------------------|
| 1. ?eḏaḥ mebrehe di?yhe ṣale | The child's tooth began to hurt. |
| 2. ləhemk di?yho mər | I touched my own stomach. |
| 3. šotede menhi mər | My stomach swelled. |
| 4. derefk di?yho ri | My head started to itch. |
| 5. qoḥom menhi ?e?ed | He scratched my hand. |
| 6. raḥaḏ šibeb di?yho fiḫa | The old man washed my clothes. |
| 7. šet daḥtar menhi naḥrer | The doctor examined my nose. |
| 8. raḥaḏk di?yho seyyara | I washed my car. |
| 9. ləhemk di?yhe ṣahan | I touched his plate. |
| 10. ?alig daḥtar menhi ?ale | The doctor cured my tooth. |
| 11. raḥaḏ daḥtar mey ri | The doctor washed his head. |
| 12. qoḥomk mey mər | I scratched his stomach. |

- (a) Translate into English: *lehēm šībēb menhi ri, raḥaḥ mebrehe diʔyhe mēr, šotede mey nāhrer, ʔeḥaḥk diʔyho mēr, raḥaḥk diʔyhe fiḫa, lehēm daḥtar diʔyhe ri.*
- (b) Translate into Soqotri, given *miʃmimo* ‘rope’, *ʔidhen* ‘ear’, *med* ‘to extend, stretch’: *The doctor cured his ear., He stretched my rope., I extended my hand., I washed his plate., The old man's ear started to itch., The child scratched my head.*

Problem 52. Quiché (2021, MSK)

Below are some sentences in Quiché and their English translations:

- | | |
|-------------------------------|-----------------------------|
| 1. Kawar lee tijoxeel. | This student sleeps. |
| 2. Katb'iinik. | You are walking. |
| 3. Oj oj ajyuq'aab'. | We are shepherds. |
| 4. Keechakun lee ajtijaab'. | These teachers are working. |
| 5. Kixwarik. | You (plural) are sleeping. |
| 6. Na kapeet ta lee elaq'oom. | This thief does not come. |
| 7. In in tijoxeel. | I am a student. |
| 8. Elaq'oom lee ajtiij. | This teacher is a thief. |

- (a) Translate into English: *Kojpeetik., At at ajtiij., Na keeb'iin ta lee elaq'omaab'.*
- (b) Translate into Quiché: *This thief is not a shepherd., I am working., These shepherds are not sleeping., This teacher is walking., You are students.*

Problem 53. Nanai (2017, MSK)

Here are some sentences in Nanai and their English translations:

- | | |
|---------------------------|------------------------------|
| 1) Arčokansal tutuguxəči. | The girls ran back. |
| 2) Wačaka dəgdəəčijni. | A heron is flying. |
| 3) Naonžokan mujkuxəni. | The boy crawled. |
| 4) Saman tutuəčixəni. | The shaman ran around. |
| 5) Gasasal dəgdəxəči. | The ducks flew. |
| 6) Moren pukčigujni. | The horse is galloping back. |
| 7) Endasal pukčixəči. | The dogs ran. |
| 8) Ambansal mujkujči. | The tigers are crawling. |

- (a) Translate into English: *Arčokan tutuxəni., Wačakasal dəgdəjči., Samansal tutugu-jči., Arčokan mujkuəčixəni.*
- (b) Translate into Nanai: *Shamans flew., Horses are galloping., The tiger ran back., Boys are running.* If multiple translations are possible, give all.

Problem 54. Yaki (2017, MSK)

Here are some sentences in Yaki and their English translations:

- U chu'u kochokame bwite. — The dog that was sleeping is running.
 - U jamut tajo'ota etbwakame kocho. — The woman who stole the clothes is sleeping.
 - U o'ou mesata jinume uka jamutta bicha. — The man who is buying a table sees the woman.
 - U chu'u uka o'outa bwikakameta ke'ek. — The dog bit the man who was singing.
 - U goi uka uusita bicha'u uka kaba'ita ke'e. — The coyote that the child sees is biting the horse.
 - U kaba'i mesata uka o'outa jinuka'u koktak. — The horse broke the table that the man bought.
- (a) Translate into English: U chu'u bwiteme uka goita bichak., U kaba'i uka chu'uta tajo'ota etbwameta ke'ek., U o'ou uka chu'uta bichaka'u tajo'ota jinu.
- (b) Translate into Yaki: *The coyote that the horse bit ran away., The man stole the dog that is singing., The dog saw the clothes that the woman is buying.* If multiple translations are possible, give all.

Problem 55. Halkomelem (2015, MSK)

Here are some sentences in Halkomelem with their English translations:

- | | |
|---|-----------------------------|
| 1. ni? q'waqwəθam'šəs. | He hit me. |
| 2. ni? č q'waqwəθam'š ʔə kwθə ʃəl'tən. | You hit me with a pencil. |
| 3. ni? ʔiləqətəs kwθə šəptən. | He bought a knife. |
| 4. ni? č ʔic'ət kwθə səplil ʔə kwθə šəptən. | You cut bread with a knife. |
| 5. ni? cən ʃəl'ət kwθə pipə. | I wrote a letter. |
| 6. ni? cən pənəlcθamə ʔə kwθə sqewθ. | I planted potatoes for you. |
| 7. ni? q'wələlcθam'šəs ʔə kwθə sce:ʔtən. | He fried salmon for me. |
| 8. ni? č ʃəl'əlcət ʔə kwθə pipə. | You wrote a letter for him. |

After studying sentences 1–8, a linguist showed a Halkomelem speaker the sentence *ni? q'waqwəθaməs*. 'He hit you'. The speaker said the sentence was incorrect, and the only way to convey this meaning was *ni? q'waqwəθa:m*. After thinking, the linguist said: "Then this sentence must also be incorrect: *ni? q'wələlcθaməs ʔə kwθə sce:ʔtən*." A native speaker confirmed the guess.

- (a) Correct sentence 11 (*ni? q'wələlcθaməs ʔə kwθə sce:ʔtən*.) and translate it into English.
- (b) Translate into English: *ni? č q'wələt kwθə sqewθ*., *ni? ʔiləqəlcətəs ʔə kwθə šəptən*..
- (c) Translate into Halkomelem: *I bought it., He planted potatoes., You cut bread for me., He wrote a letter in pencil.*

Problem 56. Catalan (2015, MSK)

Catalan verb forms and their English translations are given:

Catalan	English	Catalan	English
comprèn-ne	understand this	estimar-los	love them
comprendre'm	understand me	lleva'm	take me away
defendre'ls	defend them	m'afaita	he shaves me
els afaito	I shave them	oblidar	to forget
els defenc	I defend them	ofendre't	offend you
em pentina	he combs me	pentina'ls	comb them
en comprèn	he understands this	pentinar-me	to comb me
ens comprèn	he understands us	pentinar-te	to comb you
ens oblida	he forgets us	t'ofèn	he offends you
ens pentina	he combs us	?	he shaves us
comprèn	understand! / (2) ?	?	forget me
	?	?	forgets me
comprenc	?	?	I love this
defendre'ns	?	?	to comb us
t'estimo	?	?	I offend them
?	shave us	?	to understand this
?	he understands you		

(a) Fill in the blanks.

Problem 57. Hungarian (2015, MSK)

Here are some Hungarian verb forms and English translations in random order:

Hungarian: áldoz, csuklasz, hordasz, hordanak, hajtotok, húznak, játszol, játszotok, olvasnak, olvasol, olvastok, togyog, togyognak, togyogsz

English: you play, you hiccup, you hobble, you wear, you read, he sacrifices, he hobbles, you (pl.) play, you (pl.) bend, you (pl.) read, they hobble, they wear, they pull, they read

(a) Translate into English: áldozol, csuklotok, hajt, játszanak.

(b) Translate into Hungarian: you bend, you pull, he carries, you (pl.) sacrifice, they bend.

Problem 58. Igbo (2015, MSK)

Here are some sentences in Igbo and their English translations:

Obi gara ahịa.	Obi went to the market.
Obi agaghị ụlọ.	Obi did not go into the house.
Ike na agba egwu.	Ike is dancing.
Ike agaghị agba egwu.	Ike will not dance.
Obi zuru ụlọ.	Obi bought a house.
Obi azughị akwụkwọ.	Obi did not buy a book.
Ike na eghe azụ.	Ike is frying fish.
Ike ga agụ akwụkwọ.	Ike will read a book.
Obi ga eri anụ.	Obi will eat meat.
Ike agaghị eghe anụ.	Ike will not fry meat.
Obi ghere azụ.	Obi fried fish.

- (a) Translate into English: **Obi ga aga ụlọ., Ike erighị anụ., Obi na azu ụlọ., Ike anaghị eghe azụ.**
- (b) Translate into Igbo: *Ike ate meat., Obi does not go to the market., Obi danced., Obi will buy fish., Ike did not fry meat., Ike didn't read the book.*

Problem 59. Uara (2016, MSK)

Here are some sentences in Uara and their English translations:

English	Uara
A person comes.	Kabe ythorakwr.
Dogs come.	Ngatha ethorakwr.
A person sees a dog.	Kabef ngatha ymarwr.
Two fish see a dog.	Kofäyë ngatha ymarnth.
A dog sees two people.	Ngathaf kabe emarn.
A person holds fish.	Kabef kofä efathwr.
People hold dogs.	Kabeyë ngatha efathwrth.

- (a) Translate **Ngathayë kabe emarnth.** into English in all possible ways.
- (b) Translate into Uara: *Two people come., Dogs see a person., A fish holds two dogs.*

Problem 60. Czech (2016, MSK)

Below are some Czech phrases and their English translations:

Czech	English
abych si oblékl kabát	so that I would put on a coat
kdybychom tě zuli	if we were to take your shoes off
umyli by si ruce	they would wash their hands
aby mě oblékl	so that he would dress me
svlékli bychom ti kabát	we would take the coat off you
aby ti obuli boty	so that they would put shoes on you
kdybych se umyl	if I were to wash myself

- (a) Translate into English: **aby se zul.**

- (b) Translate into Czech: *if they would take the shoes off me, so that we would wash your hands, he would undress, I would put on shoes.*

Problem 61. Lutos (ONL- Romania, 2022)

Given below are the positive multiples of 9 that are smaller than 100, written in the Lutos language in random order.

6u zo m̄bakadε	unda so zo 6u
kal 6u	unda so zo 6u zo kal 6u
unda doi zo 6u zo mi zo doi	unda so zo doi
unda doi zo kal m̄bakadε	unda zio zo 6u zo so
unda muta zo 6u zo zio	unda zio zo mi
unda muta zo muta	

- (a) Determine the correct correspondences.
- (b) Write the following equations using arabic numerals:
- $\text{m̄bakad}\epsilon \times \text{unda doi zo mi} = \text{m̄bambo zio}$
 - $\text{unda so zo doi} + \text{unda muta zo 6u zo kal 6u} = \text{m̄bambo doi zo unda muta}$
- (c) Write in Lutos: 16, 20, 39, 46, 100, 363, 477, 800.

⚠ The Lutos language is a part of the Bongo-Bagirmi branch of the Central Sudanic languages. It is spoken by about 19 000 people in the Central African republic and Chad. 6, d, and m̄b are consonants. ε is a vowel. The language also presents two tones, however they have not been included in this problem for the sake of simplicity

Problem 62. Sulka (IOL, 2009)

The following are phrases in Sulka and their translations into the English language:

Sulka	English Translation
a vhoi a tgiang	1 betel nut
a tu a tgiang	1 yam
a lo vhoi a lomin	2 betel nuts
a lo ksie a lomin	2 coconuts
o vuo a korlotge	3 betel nuts
a moulang hori orom a tgiang	3 breadfruits
o sngu a korlolo	4 yams
o sngu a kti ^{ik} hori orom a tgiang	6 yams
o vuo a kti ^{ik} hori orom a lomin	7 betel nuts
a lo ngaitegaap hori orom a moulang	10 breadfruits
a lo ngausmia hori orom a lomin	10 coconuts
o sngu a lo kti ^{ik}	10 yams
o ngausmia a korlotge hori orom a korlotge	15 coconuts
o ngausmia a korlolo	16 coconuts
o ngaitegaap a korlolo hori orom a moulang	18 breadfruits
o sngu a lo kti ^{ik} hori orom a kti ^{ik} hori orom a korlotge	18 yams
o vuo a lo kti ^{ik} hori orom a kti ^{ik} hori orom a korlolo	19 betel nuts
o sngu a mhelom	20 yams

(a) Translate into English:

ksie a tgiang	o ngaitegaap a korlotge
o ngausmia a kti ^{ik}	o vuo a lo kti ^{ik} hori orom a tgiang

(b) Translate into Sulka

2 yams	14 yams
15 breadfruits	20 betel nuts

△ Sulka belongs to the East Papuan language family. It is spoken by approx. 3500 people in East New Britain Province in Papua New Guinea. Betel nuts are actually seeds of a certain kind of palm. Yam is the edible tuber of the tropical plant of the same name.

Problem 63. Chungli dialect of Ao (APLO, 2024)

(1) $\text{tenemri ka} + \text{metsyri asym} = \text{tenem ser metsyri pezy}$

(2) $\text{liri pezy} + \text{semyr maben pungu} = \text{tenem ser metsy maben tyko}$

- (3) $\text{tenet} + \text{ti} = \text{metsy maben pungu}$
 (4) $\text{metsy maben trok} \times \text{pungu} = \text{lir anasy}$
 (5) $\text{teri ana} \times \text{ana} + \text{rokyri asym} = \text{telang tyko maben tenet}$
 (6) $\text{semyr maben ti} + \text{liri ana} = \text{tenem ser metsy}$
 (7) $\text{tyko} \times \text{trok} = \text{tenemri pezy}$
 (8) $\text{lir anasyri pezy} - \text{lir maben pungu} = \text{tenem maben tyko}$
 (9) $\text{teri asym} \times \text{trok} = \text{lir anasy maben ti}$

- (a) Write the equalities (1–9) in digits, given that *metsy maben ti* is 18.
 (b) Write in Chungli Ao: 10, 31, 36, 58, 93. If an exact expression cannot be determined from the given data, write out all possibilities.

⚠ Chungli is one of five dialects of the Ao language, which belongs to the Sino-Tibetan family. It is spoken by approx. 130,000 people in Nagaland, India. *y* is a vowel.

Problem 64. Word Embeddings (PLO, 2020)

Word embeddings are numeric representations of words as points in d -dimensional space. That is, a word can be represented using d numbers, where each number represents the coordinate of that number along a particular dimension.

For example, 2-dimensional embeddings are representations of words as points on a plane using a pair of numbers. The first number corresponds to the x -coordinate of the point and the second number to the y -coordinate of the point.

These word embeddings are very useful representations of words for training machine learning models to perform natural language processing tasks. The embedding for a word is computed by looking at the word in a large corpus of text, and looking at what words occur in the context of that word.

Some machine translation models use embeddings of words in the source language to compute the embedding of the word in the target language, and then look up the word corresponding to that embedding. In this problem, we consider translating Hrangkhoh to English.

Here are some English words with their embeddings in the 2-D space:

<i>bull</i>	[2, 2]	<i>son-in-law</i>	[-5, 1]
<i>cow</i>	[3, 3]	<i>daughter-in-law</i>	[-4, 2]
<i>mare</i>	[4, 5]	<i>grandson</i>	[-6, 3]
<i>(female) duck</i>	[3, -1]		

Here are some Hrangkhoh words with their embeddings and their corresponding English translations in random order:

1. v $\hat{o}$ kchal [-4, 6]	A. male parrot
2. v $\hat{o}$ kpu $\hat{i}$ [-6, 8]	B. hen
3. sakorchal [-8, 6]	C. sow
4. arkhong [4, 2]	D. daughter
5. arpui [2, 4]	E. boar
6. vat $\hat{o}$ khong [4, 4]	F. granddaughter
7. vaikikhong [8, 4]	G. drake
8. naipa [-4, -12]	H. stallion
9. nainu [-6, -10]	I. son
10. tunu [-8, -10]	J. rooster

- (a) Match the Hrangkhoh words with their English translations
- (b) Write the Hrangkhoh word embeddings for the English words given with their embeddings
- (c) Write the word embeddings of the English translations A-J
- (d) Knowing that **k $\hat{e}$ l** = *goat* and **tar** = *old person* translate the following words to Hrangkhoh:
11. *female parrot*
 12. *he-goat*
 13. *old woman*

Problem 65. Lele (MSK, 2017)

Arithmetic examples are given in the Lele language*. All numbers in the examples are less than 100:

- (1) **manonoh** $\times$ **manondoruoh** = **mahangul e ondoruoh**
- (2) **manondrsih** $\times$ **manondrsih** = **manondorungul e sih**
- (3) **manonoh** $\times$ **masungul** = **manonngul**
- (4) **matoloh** + **marngul e ruoh** = **marngul e limah**
- (5) **malimngul e onoh** + **masungul e hahou** = **manondrtilngul**

- (a) Write in Lele:

2, 7, 65, 99

- (b) Write in numbers:

(6) **matilngul e toloh** + **mahahou** = **matilngul e ondrtoloh**

Problem 66. Tlapanec (Onling, 2023)

Here are some equalities in Tlapanec:

- (1) $\text{guwa-isu-mu} \times \text{ajku} = \text{ajku-rumi ijma-mo}$
- (2) $\text{mijna-guwa} \times \text{ajma} = \text{guwa-nisu isu-mu}$
- (3) $\text{mba-skiyun emba-mo} + \text{guwa-nisu ijma-mo} = \text{mba-skiyun guwa mingiyun}$
- (4) $\text{ajsu!} = \text{majju}$
- (5) $\text{mba-skiyun mijna-guwa} \times \text{ajsu} = \text{mba-jku-skiyun juan}$
- (6) $\text{guwa-ijma-mo} \times \text{ajsu} = \text{mba-skiyun guwa majju}$
- (7) $\text{ajku} \times \text{ajku} = \text{guwa-nisu emba-mo}$
- (8) $\text{mba-skiyun guwa-isu-mu} + [A] = \text{mba-jma-skiyun ijku-mu}$
- (9) $\text{mba-skiyun wisu} + [B] = \text{mba-skiyun guwa wisu}$
- (10) $[C] \times \text{ajma} = \text{mba-jsu-skiyun}$

- (a) Write the equalities 1-10 in Arabic numerals
- (b) Write the Tlapanec numerals corresponding to A, B, and C.
- (c) Write in Tlapanec:

19, 53, 69, 77, 88

Problem 67. Tifal (2017, MSK)

Arithmetic examples are given in which the numbers are replaced by Tifal numerals (none exceeds 30):

- (1) $\text{asumano} \times \text{aleeb} = \text{bokob}$
- (2) $\text{asumano} \times \text{ataling} = \text{tadang}$
- (3) $\text{bokob} \times \text{ataling} = \text{ataling madi}$
- (4) $\text{bokob} \times \text{asumano} = \text{nakal madi}$
- (5) $\text{asumano} \times \text{feet} = \text{feet madi}$
- (6) $\text{ataling} \times \text{ataling} = \text{tadang madi}$
- (7) $\text{asumano} + \text{ataling} = \text{feet}$
- (8) $\text{feet} + \text{miit} = \text{feet madi}$
- (9) $\text{tadang} + \text{ataling} = \text{tadang madi}$

- (a) Write in digits:

- (10) $\text{beeti} + \text{nakal} = \text{beeti madi}$
- (11) $\text{bokob} + \text{maakob} = \text{feet}$
- (12) $\text{awok} \times \text{awok} = \text{asumano madi}$

(b) Write the answers in Tifal:

(13) $\text{tadang} + \text{miit} = ?$

(14) $\text{ataling madi} - \text{aleeb} = ?$

Problem 68. Malayalam (2022, MSK)

Given are some numerals (in Latin transcription) from texts in classical-period Malayalam, with all their numerical values in random order:

Malayalam: nālēlu , īronpatu , īreṭṭu , īrāru , mūvāru , mūvēlu , mūvonpatu

Values: 12, 16, 18, 21, 27, 28

- (a) Establish the correct correspondences, given that the numeral mūvāru denotes an even number.
- (b) Write the value of mūvañcu , given that it denotes an odd number.
- (c) Write in Malayalam: 14, 24. If two options are possible, give both.

Problem 69. Northern Sami (2017, MSK)

The time-of-day designations in Northern Sami are given:

- 15.40 $\text{Diibmu lea logi badjel beal njeallje.}$
- 16.50 $\text{Diibmu lea logi váile vihtta.}$
- 13.10 $\text{Diibmu lea logi badjel okta.}$
- 21.25 $\text{Diibmu lea vihtta váile beal logi.}$
- 19.05 $\text{Diibmu lea vihtta badjel čieža.}$
- 12.30 $\text{Diibmu lea beal okta.}$

- (a) Write in Northern Sami: 15.55, 16.20, 18.35, 22.10.

Problem 70. Diophantine (2015, MSK)

The polynomials are given in modern notation and in Diophantine notation:

Modern	Diophantine
$40x^3 + 730x^2 + 8$	$\mathbf{K}^\gamma \mu \Delta^\gamma \psi \lambda \mathbf{M} \eta$
$40x^3 + 730x^2 + 8x$	$\mathbf{K}^\gamma \mu \Delta^\gamma \psi \lambda \zeta \eta$
$700x^6 - 660x^2 + 89x - 4$	$\mathbf{K}^\gamma \mathbf{K} \psi \zeta \pi \theta \Delta^\gamma \chi \xi \mathbf{M} \delta$
$-20x^5 + 505x^3 + 4x$	$\mathbf{K}^\gamma \varphi \varepsilon \zeta \delta \mathbf{K} \Delta^\gamma \kappa$
$-46x^5 - 28x^2 + 55$	$\mathbf{M} \nu \varepsilon \mathbf{K} \Delta^\gamma \mu \varsigma \Delta^\gamma \kappa \eta$

- (a) Write in modern notation: $\Delta^\gamma \Delta \kappa K \Delta^\gamma \lambda K^\gamma \lambda \theta$ and $K^\gamma K \mu K^\gamma \varphi \eta \Delta^\gamma \kappa \zeta \alpha$.

- (b) Write in Diophantine notation: $30x^6 + 48x^2 + 702x$; $x^3 + 81x^2 - 666$; $-500x^5 + 69x^3 - 5$.

⚠ Diophantus of Alexandria (3rd c. CE) developed an early symbolic algebra. The notation here uses Greek letters for digits and powers of the unknown.

Problem 71. Weardale Sheep Counting (2014, MSK)

The following are alphabetic notations of the prime numbers not exceeding 20 in the sheep-counting system of the Weardale Valley, County Durham:

methera-tic-a-bub teyan-a-tic tethera teyan-tic-a-bub
tethera-bub tic teyan yan-a-bub

- (a) Establish the values of all given numerals.
- (b) Write in digits: tethera-tic, methera-bub, tic-a-bub.
- (c) Write down the designations of all perfect squares not exceeding 20 in this system.

Problem 72. Finnish (Ordinals) (2018, MSK)

Here are some (now-obsolete) ordinals in Finnish and their values in random order:

Finnish: kolmasneljättä, viidesviidettä, neljäskahdeksatta, seitsemäskolmatta, viidesneljättä

Values: 27th, 33rd, 35th, 45th, 74th

- (a) Determine the correct correspondences.
- (b) Write in Finnish: 23rd, 68th.

Problem 73. Ancient Rome (2016, MSK)

Arithmetic examples are given in which numbers are written as they were sometimes written in Ancient Rome:

• + •••• = •••••
 ••• + ••••• = S••
 ••• × •••• = •
 •••• + ••••• = ?
 S × •••• = ?
 •• : ••• = ?

- (a) Fill in the blanks. Explain your solution.
- (b) In the following example, the arithmetic operation is missing rather than a number. Fill it in, knowing that it can be done in two ways: $S ? S = I$.

(b) Translate into English: акулакцэа цшындажэи хдык.

Problem 75. Thai (Time) (2020, MSK)

Time readings are given in digits and in Thai (Latin transcription) in a random order:

Digital	Thai (Latin)
2:30	(a) sam thum sip-sam
19:47	(b) ha mong chao si-sip ha
13:16	(c) ti song khrueng
21:13	(d) nueng thum si-sip chet
7:26	(e) bai ha mong sam-sip kao
11:45	(f) ti si paet
17:39	(g) nueng mong chao yi-sip hok
4:08	(h) bai nueng mong sip-hok

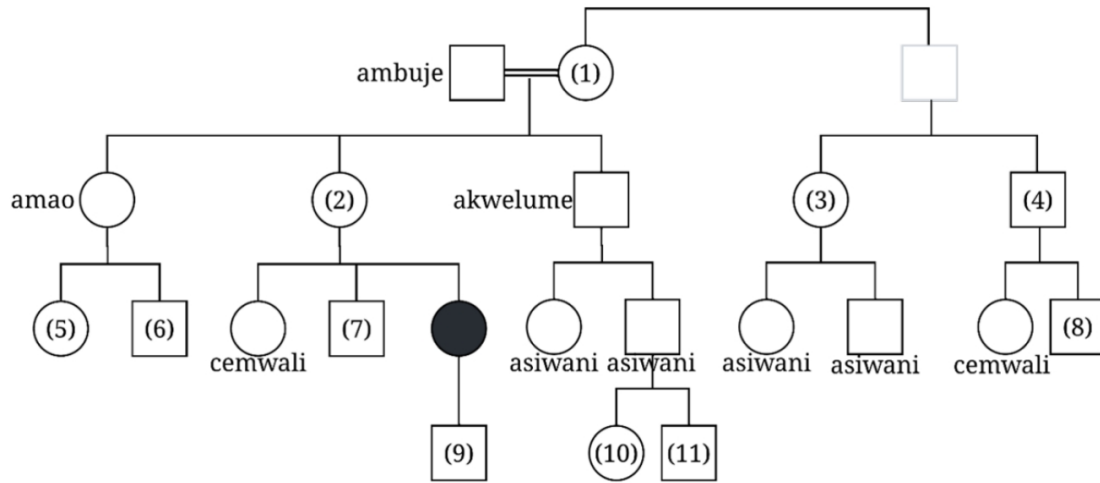
- (a) Determine the correct correspondences.
- (b) Write the digital times for: sam mong chao sip-song, ti nueng si-sip hok, bai si mong sam-sip song.
- (c) Write in Thai: 20:30, 5:28, 15:17.

Problem 76. Yao (HKLO, 2023)

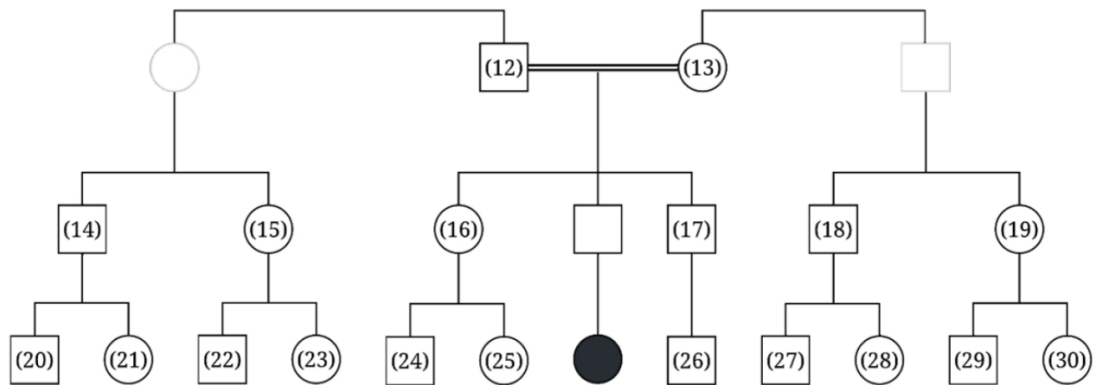
Reading a book about Yao people and the Yao language, you stumble across a chapter explaining the different meanings of kinship terms with other kinship terms:

Kinshp term	Description
ambuje	atati jua amao amao jua atati atati jua atati amao jua amao
atati	acimweni jua atati mwanache wandu jua akwelum jua amao mwanache wandu jua atati wakongwe jua amao
atati wakongwe	cemwali jua atati mwanache wakongwe jua akwelum jua amao mwanache wakongwe jua atati wakongwe jua amao
amao	cemwali jua amao mwanache wakongwe jua atati wakongwe jua atati mwanache wakongwe jua akwelum jua atati
akwelum	acimweni jua amao mwanache wandu jua atati wakongwe jua atati mwanache wandu jua akwelum jua atati
acimweni	mwanache wandu jua amao mwanache wandu jua atati
cemwali	mwanache wakongwe jua atati mwanache wakongwe jua amao
asiwani	mwanache jua atati wakongwe mwanache jua akwelum
mwanache	mwanache jua mwanache wandu jua akwelum mwanache jua mwanache wandu jua acimweni
yisikulu	mwanache jua mwanache

At the same time, you find a diagram showing a section of the family tree of a girl named Yohanna (shaded circle in the figure below). The name under each person represents the kinship term she uses to refer to them. Some words are missing:



- (a) Fill in blanks (1)-(11).
 (b) Fill in blanks (12)-(30) below.



Problem 77. Prakrit (IOLC, 2023)

Here are some Prakrit words in the Karoshti script, the latin transliteration, and the English translation, with some words missing.

Kharoṣṭhi	Transliteration	Translation
𑖦𑖩𑖪𑖫	aṽanaṃmi	in Ávana (<i>river</i>)
𑖦𑖩𑖪𑖫	cojhbo	cozbo (<i>title</i>)
𑖦𑖩𑖪𑖫	durbale	weak
𑖦𑖩𑖪𑖫	ḍhapḡeya	Ḍhapḡeya (<i>name</i>)
𑖦𑖩𑖪𑖫	pr̥theṣu	at the back
𑖦𑖩𑖪𑖫	tasmartha	for that reason
𑖦𑖩𑖪𑖫	(1)	mare given as payment
𑖦𑖩𑖪𑖫	(2)	he said
𑖦𑖩𑖪𑖫	(3)	he died
𑖦𑖩𑖪𑖫	(4)	they will do
𑖦𑖩𑖪𑖫	(5)	he dies
𑖦𑖩𑖪𑖫	(6)	to stay
𑖦𑖩𑖪𑖫	(7)	(8)
𑖦𑖩𑖪𑖫	(9)	(10)
(11)	agacham̐ti	they come
(12)	biṃnita	he broke
(13)	cmaka	Cmaka (<i>name</i>)
(14)	ichati	he wants
(15)	jheniḡa	under the care of
(16)	kariṣyatu	you (sg) will do
(17)	kartavya	to be done
(18)	ṛna	debt
(19)	(20)	they want

(a) Fill in the blanks (1)-(20).

△ Prakrit is an extinct Indo-Aryan language which was spoken around 2 thousand years ago in the present-day India. ch, ḍh, jh, m, ṣ, th, ṭh and y are consonants; ṛ is a vowel. A mark above or below a letter denotes a small change in the sound quality.

Problem 78. Buginese (2022, MSK)

Some Buginese words are given in the traditional Lontara script and in Latin transcription:

Lontara	Latin
o~w~w~	soloq
~w~w~	loka
?	kaq
^~w~	tawa
?	awu awu
<~w~	déq
?	usesseq
?	ikkoq
<~w~v~w~ ^~w~<~w~	wékkadua tattellué
^~w~w~	tellong
?	méloq
?	méng
?	doang tasiq
~w~w~	iko
v~w~w~	mario

(a) Fill in the blanks.

⚠ Lontara is a Brahmic-derived abugida.

Problem 79. Tibetan (2019, MSK)

Here are some Sanskrit words written in the Latin transcription and the Tibetan script with their translations:

Sanskrit	Tibetan
krod ^h e	ཀྲོད་ཀྱི་
p ^h ene	ཕེན་
nṛloke	ནྟི་ལོ་ཀྱི་
hayānām	ཧཱུ་ཡ་ནཱཾ་
ḍṛḍ ^h am	དྷྲྰམ་
baṭukāya	བཱུ་ཀཱུ་
alpabudd ^h aye	ཨ་ལ་བུ་དྲུ་ཡེ་
apaṭ ^h an	ཨ་པ་པ་ན་
pibanti	ཕི་བ་ནི་
g ^h orāt	གོ་ར་
udrī	ལུ་དྲི་

(a) Fill in the blanks (write Tibetan script):

b ^h rātṛn	?
k ^h e	?
ākriḍe	?
yat ^h ākḷpti	?
indreṇa	?

If you feel you can't write something, explain why.

(b) Fill in the blanks (write Latin transcription):

? ལྷ་མོ་རྒྱལ་
 ? ལྷ་མོ་རྒྱལ་
 ? ལྷ་མོ་

Problem 80. Phoenician (2016, MSK)

The Phoenician names of ten ancient cities of the Near East are given, written in traditional Phoenician script:

A	⊕ 7 4 3	F	✕ 5 7 4
B	4 7 8	G	7 4 0
C	3 7 4 4	H	4 0 4
D	Y 4 1 3	I	W 4 0
E	4 3 4 0	J	✕ 1 4 4

The names of the same cities are also given in the accepted English transmission (in random order), with the ancient names of two of the cities replaced by current ones:

Aynok, Beirut, Kadesh, Megiddo, Palmyra, Ribla, Sarepta, Tripoli, Aleppo, Ebla

- (a) Establish the correct correspondences between the Phoenician and English names for the eight cities where this is possible.
- (b) Indicate approximately what the ancient names sounded like for the two cities given by their current names.

Problem 81. Thai (Months) (2018, MSK)

The names of the months are given in Thai. The names are listed in order, but not starting with January:

1. สิงหาคม 2. กันยายน 3. ตุลาคม 4. พฤศจิกายน
5. ธันวาคม 6. มกราคม 7. กุมภาพันธ์ 8. มีนาคม
9. เมษายน 10. พฤษภาคม 11. มิถุนายน 12. กรกฎาคม

- (a) Determine which month the list starts with.